MSc Finance 20127

Corporate Finance (Business Valuation)
Cost of Capital — Part 1
Risk-Free Rate and Risk Premia

Stefano Rossi

Bocconi, IGIER, CEPR, and ECGI

Università Bocconi
DEPARTMENT OF FINANCE



Business Valuation Methods

Università Bocconi
DEPARTMENT OF FINANCE



Business Valuation Methods: DCF Models

- DCF models are based on the present value approach: The value of any asset is the present value of expected future cash flows on it

$$V_t = \sum_{s=t+1}^{n} \frac{\mathbb{E}_t\left[\mathrm{FCF}_s\right]}{(1+r)^{s-t}}$$

where:

- n: expected life of the assets (usually $n = \infty$ business)
- FCFs: Expected free cash flow in period s
- r: Discount rate reflecting the riskiness of FCFs, aka cost of capital

Università Bocconi
DEPARTMENT OF FINANCE

Cost of Capital

Università Bocconi
DEPARTMENT OF FINANCE



Cost of Capital: General Setup for Valuation

- Need to know which **discount rate** to use to compute the present value of the expected cash flows generated by the assets (or a firm's equity)
- Such discount rate is the **opportunity cost of capital**, the rate required by an investor to "hold" the expected cash flows generated by its assets
- The required return for holding the whole future cash flows of a firm is its **Weighted Average Cost of Capital or WACC** (see further)
- Why? Remember Modigliani-Miller
  - If you buy all the securities issued by a firm (debt, equity, and any other security), you have a claim on 100% of the firm's assets and thus on the total (free) cash flows
  - You also own a portfolio whose expected return is the weighted average of the expected returns on those securities, weighted by their proportion in the firm's capital structure



Weighted Average Cost of Capital (WACC)

Cost of Capital: WACC

$$\text{WACC} = r_d \cdot (1 - \tau_C) \cdot \frac{D}{(D+E+S)} + r_e \cdot \frac{E}{(D+E+S)} + r_s \cdot \frac{S}{(D+E+S)}$$

where:

- r_d: Expected Return on Debt
- r_e: Expected Return on Equity
- r_S: Expected Return on Security S (if any)
- τ_C: Corporate Income Tax Rate
- D: Market Value of Debt
- E: Market Value of Equity
- S: Market Value of Security S (if any)

Warning: using the WACC implies making specific assumptions, which are not always warranted. In such cases, you cannot use the WACC (more on this later)

Cost of Capital: Risk and Return Models

- Leading models of risk and return derive a security's expected return as the sum of the **risk free rate**, a rate investors can make on riskless investments, and the **risk premium(s)** they should require for holding risky assets

Capital Asset Pricing Model $\Longrightarrow$ $\mathbb{E}[r_e] = r_f + \beta \cdot \mathbb{E}[r_m - r_f]$

- Only one source of risk, the market portfolio, whose expected return is $\mathbb{E}[r_m]$
- β expresses the exposure to such risk

Multifactor Models $\Longrightarrow$ $\mathbb{E}[r_e] = r_f + \sum_{i=1}^{n} \beta_i \cdot \mathbb{E}[r_i - r_f]$

- Multiple sources of risks, i, whose expected returns are $\mathbb{E}[r_i]$
- β_is express the exposures to such risks

Università Bocconi
DEPARTMENT OF FINANCE

Estimating the Risk Free Rate

Università Bocconi
DEPARTMENT OF FINANCE



Estimating the Risk Free Rate: Principles

On a risk free asset, expected return = actual return. A risk free investment has:

- No default risk
- No reinvestment (rollover) risk

1. **Time horizon**
   - At any point in time, borrowers face different interest rates for cash flows of different maturities ("term structure" of interest rates)
   - Given a term structure of (risk free) government bond rates, need to match the maturity of the asset being evaluated with the maturity of the (risk free) government bond rate

2. **Not all government securities are risk-free**
   - Some governments face default risk $\Longrightarrow$ rates on their bonds will not be risk-free
   - This is true also if they borrow in a currency other than their own

3. **Consistency principle**
   - Nominal cash flows require nominal risk free rates
   - The currency of the cash flows must be the currency of the risk free rate

Estimating the Risk Free Rate: Real vs Nominal Rates

- Under high and unstable inflation, valuation is often done in real terms $\Longrightarrow$ make sure cash flows reflect real growth rates and not growth coming from price inflation.

- How to estimate real expected rate of return?

1. **"Subtracting" inflation rate (standard approach)**:

   r_f real $\approx$ r_f nominal $-$ expected inflation (i) $\longrightarrow$ holds only for low i

   r_f real $= \frac{1+r_f \text{ nominal}}{1+i} - 1$

2. **Use inflation indexed Treasuries (TIPs)**:

   TIPs offer a guaranteed real return $\longrightarrow$ only for US (low and stable i)

3. **No financial markets frictions (Capitals can flow freely)**:

   No differences in real r_f across countries $\longrightarrow$ use TIPs for all countries

4. **Financial markets frictions**:

   Expected long term real r_f should equal expected long term growth (g)

Estimating the Risk Free Rate: Sovereign Default Spreads

1. **Local Currency Government Bond**:
   (a) **Default spread approach**

   $$r_f \text{ in Local Currency (LC)} = \text{Govt Bond rate (LC)} - \text{Default Spread}$$

   - Default spread come from USD credit spread (see next)
   - For countries that do not issue USD bonds, use average spread for countries in same rating class

   (b) **Insurance (Credit Default Spread – CDS) approach**

   $$r_f \text{ in Local Currency (LC)} = \text{Govt Bond rate (LC)} - \text{CDS Spread}$$

2. **Build-up approach** :
   - If there is no traded government bond in local foreign currency:

   $$\text{Build-up } r_f = \text{Expected inflation + Expected } 1/t \text{ real growth rate}$$

3. **Derivatives Markets (no frictions in capital markets)**:
   - Use Covered Interest Parity (CIP) to back out r_f in Local Currency (LC):

   $$\text{Forward rate}_{LC,USD} = \text{Spot rate}_{LC,USD} \times \frac{1+r_{f,LC}}{1+r_{f,USD}}$$

4. **Risk-free Rate Conversion (frictions in capital markets)**:
   - Use Uncovered Interest Parity to back out r_f LC:

   $$(1 + r_{f,USD}) = (1 + r_{f,LC}) \frac{1+\text{Expected Inflation}_{USD}}{1+\text{Expected Inflation}_{LC}}$$

Estimating the Risk Free Rate: Bond Rates (January 2022)

Currency	Govt Bond Rate	Currency	Govt Bond Rate	Currency	Govt Bond Rate
Australian $	1.67%	Indian Rupee	6.45%	Qatari Dinar	2.11%
Brazilian Reai	10.31%	Indonesian Rupiah	1.02%	Romanian Lev	5.21%
British Pound	0.97%	Israeli Shekel	1.02%	Russian Ruble	8.42%
Bulgarian Lev	0.70%	Japanese Yen	0.07%	Singapore $	1.67%
Canadian $	1.43%	Kenyan Shilling	12.55%	South African Rand	9.35%
Chilean Peso	5.72%	Korean Won	2.26%	Swedish Krona	0.21%
Chinese Yuan	2.85%	Malyasian Ringgit	3.59%	Swiss Franc	-0.14%
Colombian Peso	8.14%	Mexican Peso	7.56%	Taiwanese $	0.69%
Croatian Kuna	0.63%	Nigerian Naira	12.61%	Thai Baht	1.95%
Czech Koruna	2.98%	Norwegian Krone	1.72%	Turkish Lira	23.33%
Danish Krone	0.10%	NZ $	2.33%	US $	1.51%
Euro	-0.18%	Pakistani Rupee	11.54%	Vietnamese Dong	2.12%
HK $	1.45%	Peruvian Sol	5.87%	Zambian kwacha	31.50%
Hungarian Forint	4.58%	Phillipine Peso	4.90%		
Iceland Krona	4.15%	Polish Zloty	3.66%		

Università Bocconi
DEPARTMENT OF FINANCE

Estimating the Risk Free Rate: Bond Rates (January 2022)

Country	$ Bond Rate	Riskfree Rate	Default Spread
	$ Bonds		
Peru	3.66%	1.51%	2.15%
Brazil	3.70%	1.51%	2.19%
Colombia	2.33%	1.51%	0.82%
Poland	1.80%	1.51%	0.29%
Turkey	5.43%	1.51%	3.92%
Mexico	2.01%	1.51%	0.50%
Russia	2.48%	1.51%	0.97%
	Euro Bonds		
Bulgaria	1.00%	-0.18%	1.18%

Estimating the Risk Free Rate: CDS Spreads (January 2022)

Country	1/1/22	CDS Spread net of US	Country	1/1/22	CDS Spread net of US	Country	1/1/22	CDS Spread net of US
Abu Dhabi	0.77%	0.58%	Greece	1.69%	1.50%	Panama	1.26%	1.07%
Algeria	1.10%	0.91%	Guatamela	2.06%	1.87%	Peru	1.31%	1.12%
Angola	5.94%	5.75%	Hong Kong	0.41%	0.22%	Philippines	0.92%	0.73%
Argentina	23.32%	23.13%	Hungary	0.69%	0.50%	Poland	0.68%	0.49%
Australia	0.23%	0.04%	Iceland	0.73%	0.54%	Portugal	0.56%	0.37%
Austria	0.19%	0.00%	India	1.44%	1.25%	Qatar	0.74%	0.55%
Bahrain	3.40%	3.21%	Indonesia	1.36%	1.17%	Romania	1.24%	1.05%
Belgium	0.21%	0.02%	Iraq	5.63%	5.44%	Russia	1.70%	1.51%
Brazil	2.91%	2.72%	Ireland	0.27%	0.08%	Rwanda	3.36%	3.17%
Bulgaria	0.81%	0.62%	Israel	0.72%	0.53%	Saudi Arabia	0.88%	0.69%
Cameroon	3.56%	3.37%	Italy	1.41%	1.22%	Senegal	2.66%	2.47%
Canada	0.28%	0.09%	Japan	0.33%	0.14%	Serbia	1.37%	1.18%
Chile	1.25%	1.06%	Kazakhstan	0.99%	0.80%	Slovakia	0.63%	0.44%
China	0.74%	0.55%	Kenya	4.44%	4.25%	Slovenia	0.87%	0.68%
Colombia	2.77%	2.58%	Korea	0.35%	0.16%	South Africa	2.85%	2.66%
Costa Rica	3.92%	3.73%	Kuwait	0.86%	0.67%	Spain	0.60%	0.41%
Croatia	1.11%	0.92%	Latvia	0.74%	0.55%	Sri Lanka	19.69%	19.50%
Cyprus	0.74%	0.55%	Lebanon	NA	NA	Sweden	0.19%	0.00%
Czech Republic	0.47%	0.28%	Lithuania	0.79%	0.60%	Switzerland	0.11%	0.00%
Denmark	0.15%	0.00%	Malaysia	0.81%	0.62%	Thailand	0.52%	0.33%
Dubai	1.33%	1.14%	Mexico	1.58%	1.39%	Tunisia	8.82%	8.63%
Ecuador	7.57%	7.38%	Morocco	1.32%	1.13%	Turkey	5.51%	5.32%
Egypt	5.74%	5.55%	Netherlands	0.19%	0.00%	Ukraine	6.17%	5.98%
El Salvador	18.33%	18.14%	New Zealand	0.21%	0.02%	United Kingdom	0.18%	0.00%
Estonia	0.85%	0.66%	Nicaragua	4.36%	4.17%	United States	0.19%	0.00%
Ethiopia	20.40%	20.21%	Nigeria	5.53%	5.34%	Uruguay	1.46%	1.27%
Finland	0.20%	0.01%	Norway	0.19%	0.00%	Venezuela	NA	NA
France	0.34%	0.15%	Oman	3.19%	3.00%	Vietnam	1.56%	1.37%
Germany	0.18%	0.00%	Pakistan	3.67%	3.48%	Zambia	NA	NA

Università Bocconi
DEPARTMENT OF FINANCE

Estimating the Risk Free Rate: Bond Ratings (January 2022)

S&P Sovereign Rating	*Moody's Sovereign Rating*	*Default Spread*
AAA	Aaa	0.00%
AA+	Aa1	0.34%
AA	Aa2	0.42%
AA-	Aa3	0.51%
A+	A1	0.60%
A	A2	0.72%
A-	A3	1.02%
BBB+	Baa1	1.36%
BBB	Baa2	1.62%
BBB-	Baa3	1.87%
BB+	Ba1	2.13%
BB	Ba2	2.56%
BB	Ba3	3.06%
B+	B1	3.83%
B	B2	4.68%
B-	B3	5.53%
CCC+	Caa1	6.38%
CCC	Caa2	7.66%
CCC-	Caa3	8.51%
CC+	Ca1	10.21%
CC	Ca2	12.88%
CC-	Ca3	14.50%
C+	C1	15.50%
C	C2	17.00%
C-	C3	19.00%

Test 3: Risk Free Rate in a Currency?

The Brazilian government bond rate in nominal Real on January 1st, 2022 was 10.31%.
To get to a r_f in nominal Real, we can use one of three approaches

1. **Government Bond Spread**:
   - The 2032 USD-denominated Brazilian bond has a spread of 2.19% over the US T-Bond rate
     r_f in LC = 10.31% − 2.19% = 8.12%

2. **CDS Spread**:
   - The CDS spread for Brazil, adjusted for the USD CDS spread was 2.72%
     r_f in LC = 10.31% − 2.72% = 7.59%

3. **Rating-based Spread**:
   - Brazil had a Ba2 Moody's rating in LC; the default spread for that class is 2.56%
     r_f in LC = 10.31% − 2.56% = 7.75%

Estimating the Equity Risk Premium (ERP)

Università Bocconi
DEPARTMENT OF FINANCE



Estimating the Equity Risk Premium: Definitions

- ERP: the premium a well diversified *marginal* investor requires to hold the market portfolio rather than a risk free asset
- Historical ERP: the premium that stocks have historically earned over riskless securities over long time periods.
  - Implicit assumption: history can forecast the future
- Implied ERP: the forward looking premium that is implicitly priced in the current stock market level, given expected aggregate cash flows and risk free rates[1]
  - Implicit assumption: the market, overall and on average, is correctly priced

Università Bocconi
DEPARTMENT OF FINANCE

(1): See Martin, I., 2017, What is the expected return on the market?, *Quarterly Journal of Economics* 132(1), 367-433

Estimating the Equity Risk Premium: Peso Problems

- Question: What about rare shocks??
  - Also called Peso problems or dark swans
  - Very important, not only in academia but also among practitioners

- Answer: At most 1-2% of historical equity premium. Not unimportant. Insurable with index options

  But other long-run decline problems are harder to insure

Estimating the Equity Risk Premium: HRP Approach (US) — I

Estimates of Historical Risk Premiums (HRP) are sensitive to some choices

1. Time period used
   - ERP is time varying: different time periods produce different measures
   - Shorter time periods usually have higher standard errors
2. Choice of risk-free security
   - Short term (T-Bills) versus Long term (T-Bond) risk free assets
   - If the term structure is upward sloping, ERP is larger when estimated over shorter term risk free assets
   - Consistency with the risk free rate chosen to estimate the cost of capital
3. Arithmetic vs geometric average
   - Arithmetic average: when returns are serially uncorrelated and the estimate of the ERP is for the next year
   - Geometric average: when returns are serially correlated (Fama and French 1988) and it adjusts for the long term view (although we are using 1-period models to estimate the cost of capital)

Estimating the Equity Risk Premium: HRP Approach (US) — II

Even after settling on the least unbiased choices, there are caveats

1. Noisy estimates
   - Even over long time periods, the risk premium estimate will have a large standard error
   - E.g., over 1928-2019 (about 90 years), assuming a standard deviation of 20% in annual stock returns, your standard error $\geq$ 2% (Standard Error in Premium $\approx 20\%/\sqrt{90} = 2.1\%$)
   - The implied standard deviation in equities rose to almost 50% during the last quarter of 2008. Think about the consequences for HRP estimates, if this volatility persisted

2. Survivorship bias
   - Using historical data from the US equity markets over the 20th century creates a sampling bias since we look only at company which "survived"
   - US equity markets were amongst the most successful in the 20th century

$\Longrightarrow$ **There is a trade-off between standard errors and survivorship bias**

Estimating the Equity Risk Premium: HRP Approach (Other Markets)

- It is already challenging to estimate a reliable historical premium for the US; the more so for markets with shorter and more volatile histories. One solution:

Modified Historical Risk Premium

ERP = Base Premium for mature equity markets + Country Premium

1. What should the base ERP for a mature equity market be?
   - US market seems to be the best reference (available); at end of 2021 it was 4.72%
2. Should there be a Country Premium?
   - Some argue country risk is fully diversifiable (*open markets*)
   - Stulz (1999) argues country risk is not always diversifiable (*segmented markets* and/or cross-country correlation)
   - Damodaran: add Country Risk Premium (CRP) if estimate cost of capital for non-US markets
3. If we want to have a Country Premium, how do we estimate it?
   - Need to measure the **absolute country risk**
   - Need to measure the **exposure** to the country risk

Estimating the Equity Risk Premium: CRP (Default Risk Spreads)

CRP = default spread of the bond issued by the country

- But only if it is denominated in a currency for which a default free entity exists

1. Add default spread to the USD denominated expected return on equity

$$r_{e,USD} = r_f + \beta \cdot \text{ERP}_{US} + \text{Default Spread}$$

2. Add default spread to the USD denominated ERP

$$r_{e,USD} = r_f + \beta \cdot (\text{ERP}_{US} + \text{Default Spread})$$

- This approach presents the following shortcomings:
  a. Ratings agencies seem to lag markets in responding to changes in CRP
  b. Focus on default risk may obscure other risks that could affect equity markets
- Plausible solutions to the shortcomings:
  a. Use Credits Default Spreads (CDS) over a long time period (e.g. 10 year)
  b. Use other measure of risk, such as (i) insurance prices (if available), (ii) country scores (The Economist), or (iii) build-up CRP from fundamentals
  c. Correct the default spreads by the relative standard deviation (see later)

Estimating the Equity Risk Premium: CRP (Relative Std Deviation)

The CRP is based upon the volatility (σ) of the market in question relative to US market

1. Compute the relative standard deviation of equity market "X" over US equity market:

$$\text{Relative Standard Deviation}_X = \frac{\sigma_X}{\sigma_{US}}$$

2. Adjust US ERP to estimate the Country Specific ERP:

$$\begin{aligned} \text{ERP}_X &= \text{ERP}_{US} \times \text{Relative Standard Deviation}_X \\ \text{CRP}_X &= \text{ERP}_X - \text{ERP}_{US} \end{aligned}$$

3. Use the estimated ERP for country "X" to estimate r_e
   - This approach presents the following shortcomings:
     a. Markets may have different structures and liquidity affecting volatility
     b. There are risky emerging markets with low volatility driven by illiquidity
     c. Currency differences between markets (USD vs local currency)
   - Plausible solutions to the shortcomings:
     a.-b. Examine carefully the market structure and its liquidity
     c. Compute USD denominated returns when estimating volatility of market "X"

Estimating the ERP: CRP (Default Spreads & Relative Std Dev)

CRP = bond default spread $\times$ relative volatility of stock and bond prices in that market

1. Compute the relative volatility of equity over bonds markets:

$$\text{Relative Standard Deviation}_X = \frac{\sigma_{Equity,X}}{\sigma_{Bonds,X}}$$

2. Adjust the country default spread to estimate the CRP:

$$\text{CRP}_X = \text{Country Default Spread} \cdot \left(\frac{\sigma_{Equity,X}}{\sigma_{Bonds,X}} \right)$$

3. Use the estimated CRP for country "X" to estimate r_e as above:
   a. Add default spread to the USD denominated expected return on equity

$$r_{e,USD} = r_f + \beta \cdot \text{ERP}_{US} + \text{CRP}_X$$

   b. Add default spread to the USD denominated ERP

$$r_{e,USD} = r_f + \beta \cdot (\text{ERP}_{US} + \text{CRP}_X)$$

- This approach presents the following shortcomings:
  a. Similar to the default spreads and relative standard deviation approaches
- Plausible solutions to the shortcomings:
  a. Similar solutions to above

Estimating the Equity Risk Premium: CRP (Exposure)

Three possible ways to estimating CRP, given relation between ERPs in LC and USD

$$r_{e,LC} = \left[(1 + r_{e,USD}) \cdot \left(\frac{1 + i_{LC}}{1 + i_{USD}} \right) - 1 \right]$$

1. **Assuming every company in the country is equally exposed to country risk**

$$r_{e,USD} = r_f + \beta \cdot \text{ERP}_{US} + \text{CRP}_{X,USD}$$

   - Implicit when using the local government's dollar borrowing rate as risk free rate

2. **Assuming a company's exposure to country risk is same as its exposure to market risk**

$$r_{e,USD} = r_f + \beta \cdot (\text{ERP}_{US} + \text{CRP}_{X,USD})$$

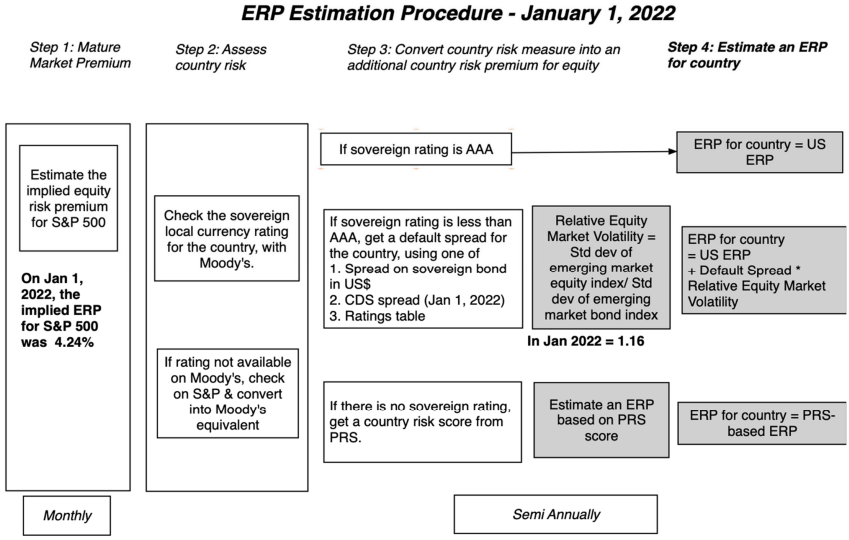
   - Implicit assumption: country risk is a portion of market risk

3. **Treat country risk as separate risk factor; allow firms to have different exposure**

$$r_{e,USD} = r_f + \beta \cdot \text{ERP}_{US} + \lambda \cdot \text{CRP}_{X,USD}$$

   - Implicit assumption: multifactor model

Estimating the Equity Risk Premium: Estimation Procedure



Estimating the Equity Risk Premium: Implied ERP (January 2022)

In 2021, earnings recovered almost entirely from the 2020 collapse and dividends & buybacks surged as well.

Base year cash flow (last 12 mths)
Dividends (TTM): 59.20
+ Buybacks (TTM): 88.05
= Cash to investors (TTM): **147.24**

Expected earnings/cashflow growth in next 5 years
Analysts were on target in 2021, with estimated earnings of 206.38 for the year. They were projecting compunded annual growth rate of 6.47% a year for next five years.

Modified Payout
This computation assumes that the payout ratio stays constant over time. If you assume that it changes to a sustainable level (given stable growth & ROE), the implied ERP = 4.90%

Actual numbers | Forecasted numbers

	2019	2020	*LTM*	*2021E*	1	2	3	4	5
Expected Earnings	157.18	139.76	190.34	206.38	219.74	233.96	249.11	265.23	282.40
Cash payout as % of earnings	146.31	118.66	77.36%	77.36%	77.36%	77.36%	77.36%	77.36%	77.36%
Dividends + Buybacks =	93.08%	84.90%	147.24	159.65	$169.99	$180.99	$192.71	$205.18	$218.46

Earnings and Cash flows grow @1.51% (set equal to risk free rate) a year forever.

S&P 500 on 1/1/22= **4766.18**

$$4766.18 = \frac{169.99}{(1+r)} + \frac{180.99}{(1+r)^2} + \frac{192.71}{(1+r)^3} + \frac{205.18}{(1+r)^4} + \frac{218.46}{(1+r)^5} + \frac{218.46\,(1.0151)}{(r-.0151)(1+r)^5}$$

The last term in this equation is the expected index level at the end of year 5 (capturing price appreciation)

Solve for r

You can solve for this r either iteratively or using the solver function in Excel

r = Implied Expected Return on Stocks = 5.75%

Minus

Risk free rate = T.Bond rate on 1/1/22= 1.51%

Equals

Implied Equity Risk Premium (1/1/22) = 5.75% - 1.51% = 4.24%

Università Bocconi
DEPARTMENT OF FINANCE

Takeaways

- The CAPM is a model of how risk is priced in capital markets
- In the CAPM, idiosyncratic risk can be diversified away $\Rightarrow$ only systematic risk is priced
- The CAPM is empirically rejected. However, it is still widely used to estimate the equity cost of capital. To do so, we need to estimate 3 parameters
  - Risk-Free rate, typically the yield of a long-term government bond
  - Beta, measured using historical monthly, weekly or daily returns
    - Estimates of beta are noisy; to improve precision of estimates, can use comparable companies
    - Beta needs to be measured over a period in which the firm's leverage and operations are stable
  - Equity Market Risk Premium (ERP), the expected return on some broad-based index in excess of the Risk-Free rate
  - The estimates of beta, ERP, and Risk-Free Rate are logically linked together

MSc Finance 20127

Corporate Finance (Business Valuation)
Cost of Capital — Part 2
Risk Parameters and Cost of Financing

Stefano Rossi

Bocconi, IGIER, CEPR, and ECGI

Università Bocconi
DEPARTMENT OF FINANCE

Estimating Beta (β)

Estimating β: Historical Market βs

- To estimate βs we regress stock returns R_i against market returns R_m:

$$r_{it} = \alpha + \beta \cdot r_{mt} + \varepsilon_{it}, \text{ with } \beta = \frac{\text{Cov}\,(r_i, r_m)}{\text{Var}\; r_m} = \frac{\sigma_{im}}{\sigma_m^2}$$

  - where α is the intercept and β is the slope of the regression, which corresponds to the beta of the stock, and measures the riskiness of the stock relative to the risk of the market

- To reconcile CAPM theory with the regression, we use Jensen's α:

$$\mathbb{E}\left[r_i\right] = r_f + \beta \cdot \mathbb{E}\left[r_m - r_f\right] = r_f \cdot (1 - \beta) + \beta \cdot \mathbb{E}\left[r_m\right]$$

  - Compare α from the regression (ex-post) with $r_f \cdot (1 - \beta)$ (ex ante):

$\alpha > r_f \cdot (1 - \beta)$	stock did better than expected, over a certain period
$\alpha = r_f \cdot (1 - \beta)$	stock did as expected, over a certain period
$\alpha < r_f \cdot (1 - \beta)$	stock did worse than expected, over a certain period

- Blume's adjustment (e.g. Bloomberg): $\beta_{Adj} = \beta_{Raw} \cdot (2/3) + 1.00 \cdot (1/3)$
- This approach presents the following shortcomings
  - High standard errors;
  - Firm's business mix and leverage over the regression period, not current ones
  - Not available for private firms / not reliable for illiquid markets

Estimating β: Historical Market βs — Summary of Problems

1. Historical returns
   - Using a longer time period allows for a greater number of observations. Longer time periods, though, might introduce biases due to structural changes in the firm/industry risk profile. This is the main reason supporting using a shorter period, between 2 and 5 years

2. Return type
   - Both stock returns and market returns should be calculated considering both the capital gains and the dividends received during the period under consideration
   - Some sources calculate returns only based on price changes without considering dividends

3. Frequency (Return Interval)
   - Using daily returns increases the number of observations in the regression. Nevertheless, such an approach exposes the estimation process to a significant bias in beta estimates caused by the lack of liquidity of certain stocks

4. The market index
   - Due to the emergence of internationally diversified investors who allocate their assets in different markets, it is increasingly common to calculate β with respect to international indices, (e.g. Morgan Stanley Capital Index)

Estimating β: Historical Market βs — Solutions

1. Modify the regression beta by
   - Changing the index used to estimate the beta
   - Changing the frequency of returns
   - Changing the length of the observation period
   - Reducing the biases (see Dimson, 1979; Scholes and Williams, 1977)

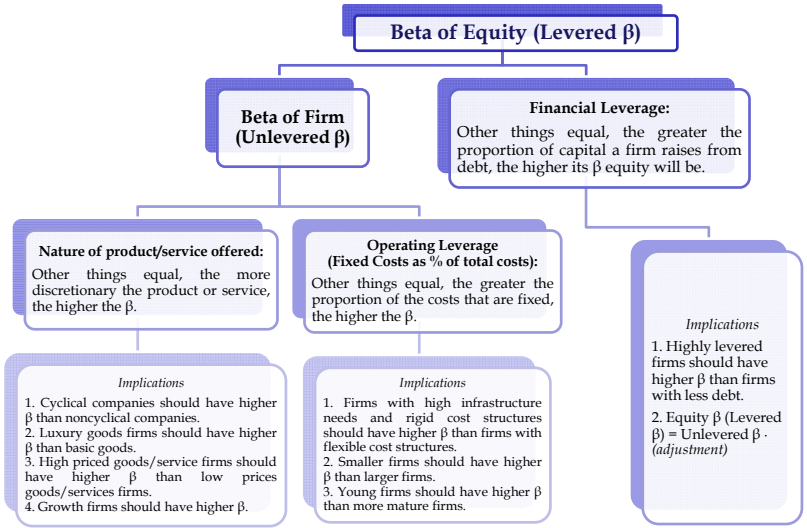
2. Estimate the beta for the firm using "Accounting βs "
   - Regress change in the firm's earnings against change in market's earnings
   - Problems: (i) earning smoothing, (ii) non-operating accounting policies, (iii) frequency of earnings release (at most, quarterly)

3. Estimate firm's β bottom-up (often preferable). This requires (more on this later):
   - Understanding the type of business / businesses the firm is in
   - Understanding the business mix of the firm
   - Estimating the financial leverage of the firm

Università Bocconi
DEPARTMENT OF FINANCE

$$\beta_P = \beta_a \cdot \omega_a + \beta_b \cdot \omega_b = \beta_a \cdot \frac{A}{A+b} + \beta_b \cdot \frac{B}{A+b} \iff \beta_P = \sum_{i=1}^{n} \beta_i \cdot \omega_i$$

Estimating β: Bottom Up βs — Determinants



Estimating β: Bottom Up βs — An Ideal World



- Two alternative approaches to unlever/relever beta, depending on assumption on debt:

1. If debt is fixed as a % of total capital ($D/(D+E)$) (debt grows as overall business grows):

$$\beta_{\text{levered}} = \beta_{\text{unlevered}} \times \left[1 + \frac{D}{E}\right]$$

2. If debt is fixed in \$ terms (as in Modigliani-Miller), we get Hamada formula:

$$\beta_{\text{levered}} = \beta_{\text{unlevered}} \times \left[1 + (1 - \tau_C) \cdot \frac{D}{E}\right]$$

Università Bocconi
DEPARTMENT OF FINANCE

Estimating β: Bottom Up βs — Real World

Step 1: Find the business/es that your firm operates in.

Step 2: Find publicly traded firms in each of these businesses and obtain their regression β. Unlever each regression β using these businesses' own D/E ratio: *Unlevered β for business = Regression β for business / (1 + (D/E ratio of business) [or Hamada Equation if using APV]*.
Compute the (simple or cap weighted) average/median of these unlevered β, which constitutes your estimate.
Relever using your firm's D/E (for pure-players only)[1].

Possible Refinements

If you can (i.e. have enough information), adjust this β for *differences* between your firm and the comparable firms on operating leverage and product characteristics.

Step 3: Estimate how much value your firm derives from each of the different businesses it is in.

While revenues or operating income are often used as weights, it is better to try to estimate the value of each business.

Step 4: Compute a weighted average of the unlevered β of the different businesses (step 2) using the weights from Step 3.
Bottom-up Unlevered β for your firm = Weighted average of the unlevered β of the individual business (*see before*)

If you expect the business mix of your firm to change over time, you can change the weights on a year-to-year basis.

Step 5: Compute a levered β (equity β) for your firm, using the market D/E ratio for your firm.
$\beta_l = \beta_u(1+ (D/E))$ *[or Hamada Equation if using APV] (*)*
(*) = [if riskless Debt – see later]

If you expect your D/E ratio to change over time, the levered β will change over time.

Estimating β: Bottom Up βs — Adjusting Financial Leverage

1. Hamada Approach

If we assume that debt carries no market risk (i.e., $\beta_D = 0$) and the amount of debt D is fixed in dollar terms, the beta of levered equity, β_L, can be written as

$$\beta_L = \beta_U \cdot \left[1 + (1 - \tau_C) \cdot \frac{D}{E}\right]$$

2. Textbook WACC Approach

If D/E is fixed and $\beta_D = 0$, use

$$\beta_L = \beta_U \cdot \left[1 + \frac{D}{E}\right]$$

3. Debt Adjusted Approach

If debt carries market risk (i.e., $\beta_D \neq 0$) and you can estimate β_D, then β_L is

$$\beta_L = \beta_U \cdot \left[1 + (1 - \tau_C) \cdot \frac{D}{E}\right] - \beta_D \cdot (1 - \tau_C) \cdot \frac{D}{E}, \text{ or}$$

$$\beta_L = \beta_U \cdot \left[1 + \frac{D}{E}\right] - \beta_D \cdot \frac{D}{E}$$

depending on whether D is fixed in dollar terms or D/E is fixed in percent terms

- While the Debt Adjusted Approach is desirable, estimating β_D is often challenging

Università Bocconi
DEPARTMENT OF FINANCE

Estimating β: Bottom Up βs — Reasons to Use It

- Lower noise

  The standard error in a bottom up β will be significantly lower than the standard error in a single regression β. Roughly speaking, the standard error of a bottom up β estimate is:

$$\text{Std Error of bottom up } \beta = \frac{\text{Average Std Error across } \beta s}{\sqrt{\#\text{ of firms in sample}}}$$

- Current Operating Structure

  The bottom up β can be adjusted to reflect changes in the firm's business mix and financial leverage. Regression β reflects the past

- Financial Structure

  The bottom up β can be adjusted to reflect recent or future changes (e.g. restructuring) in the firm's financial structure

- Private Firms / Illiquid Stocks / Start-Ups

  you can estimate bottom up β even when you do not have historical stock prices (e.g. IPO, Company's divisions, Start-Ups, ...)

Estimating Lambda (λ)

Università Bocconi
DEPARTMENT OF FINANCE



Estimating λ: Determinants

If you treat country risk as **separate risk factor** and allow firms to have different exposures

$$r_{e,USD} = r_{f,USD} + \beta \cdot \text{ERP}_{US} + \lambda_{firm,X} \cdot \text{CRP}_{X,USD}$$

$$r_{e,LC} = \left[(1 + r_{e,USD}) \cdot \left(\frac{1 + i_{LC}}{1 + i_{USD}}\right) - 1\right]$$

- **Source of revenues**
  - Other things equal, a firm should be more exposed to risk in a country if it generates more of its revenues there. A Brazilian firm that generates the bulk of its revenues in Brazil should be more exposed to country risk that one that generates a smaller % of its revenues within Brazil
- **Manufacturing facilities**
  - Other things equal, a firm that has all of its production facilities in Brazil should be more exposed to country risk than one with production facilities spread over multiple countries
  - Problem amplified for firms that cannot move their production facilities (e.g. mining and oil firms)
- **Use of risk management**
  - Companies can use options/futures markets, insurance, and real investment to hedge country risk

Estimating λ: Approaches

1. Revenue breakdown

The easiest and most accessible data is on revenues, and most companies break their revenues down by region. If so, can estimate λ as the proportion of a firm's revenues generated in a country and scale this to the proportion of domestic revenue generated by the average firm in that country

$$\lambda_{firm,X} = \frac{\%\text{ of revenues in country X}_{\text{firm}}}{\%\text{ of domestic revenues}_{\text{average firm in country X}}}$$

2. Regression vs Country Bond

Run regressions of stock returns for each firm against the returns of the country bond, for the country X we want to estimate λ. The implicit assumption is that country bonds' returns are a function of the country risk

$$\begin{aligned} r_{e,firm} &= \alpha + \lambda_{firm,X} \cdot r_{d,X} + \varepsilon \\ \lambda_{firm,X} &= \frac{\text{Cov}\left(r_{e,firm}, r_{d,X}\right)}{\text{Var } r_{d,X}} \end{aligned}$$

The Cost of Equity (k_e) — Recap

Università Bocconi
DEPARTMENT OF FINANCE



Cost of Equity: Recap

Market Risk Exposure (β)

1. Regression β
2. Accounting β
3. Bottom-up β (preferred): based upon other firms in the business, and firm's own financial leverage

Country Risk Exposure (λ)

1. λ =1
2. λ =β
3. λ estimated specifically

$$r_{e,USD} = r_{f,USD} + \beta(ERP_{US}) + \lambda_X(CRP_{X,USD})$$

Risk free rate

- has to be "fully" *risk-free*
- has to be in the same *currency* as cash flows
- has to be defined in same *terms* (real or nominal) as the cash flows

Historical Premium

1. Mature Equity Market Premium: average premium earned by stocks over T-Bonds in U.S.

2. Country risk premium:
   - default spread
   - relative standard deviation
   - default spread + relative standard dev

or

Implied Premium

Based on (i) how equity market is priced today and (ii) a simple valuation present value model, from which the implied premium is backed-out

The Cost of Debt (k_d) — Recap

Università Bocconi
DEPARTMENT OF FINANCE



Cost of Debt: Determinants

- The cost of Debt r_D is the rate at which a firm can borrow, given a certain debt maturity
- In general, it holds that:
  - Promised Interest Rate (i.e. Promised Yield) $\neq$ Expected Interest Rate
  - Promised Interest Rate $=$ Time Premium + default premium
  - Expected Interest Rate $=$ Time Premium $\Longleftarrow$ r_D ¡ Promised Yield
- Determinants of the Cost of Debt:
  1. Risk free rate:
     - Most basic component: as the risk free rate increases, the cost of Debt will also increase
  2. Default risk of the Company:
     - As default risk (and the associated default spread) increases, the Cost of Debt will also increase.
     - It is somehow related also to the Country Risk (e.g. cost of debt for Italian Companies in 2011)
     - need to be careful, because not all companies may have equal Country Risk exposure
  3. The tax shield on Debt:
     - since interests on Debt are tax deductible, the After Tax Cost of Debt is a function of the marginal corporate tax rate τ_C:

$$\text{After Tax Cost of Debt} = \text{Pre-Tax Cost of Debt} \times (1 - \tau_C)$$

Cost of Debt: Approaches I

Most widely used approaches to estimating cost of Debt:

1. **Recent borrowing history**
   - Look at yield to maturity on a liquid straight bond recently outstanding from same firm
   - Limitation: very few firms have long term straight bonds that are highly liquid and traded
2. **Public rating of the company**
   - look at the firm's rating and estimate a default spread based upon the rating
   - Limitation: while this approach is more robust than 1., different bonds from the same firm can have different ratings. If so, need to use a median rating for the firm
3. **Synthetic rating of the company**
   - If firm has no ratings or multiple ratings, estimate a synthetic rating and the cost of debt based upon it
   - Synthetic rating as a function of interest coverage ratio:

$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest Expenses}}$$

   - There exist different models to estimate synthetic ratings (e.g. Altman's Z-score)

Cost of Debt: Approaches II

- We only observe promised yield (Yield to Maturity), not the cost of debt r_D
- We can estimate the cost of debt r_D as an internal rate of return (IRR) using expected cash flows (payments) and current price

$$\frac{\mathbb{E}_0\left[\mathrm{CF}_1\right]}{(1+r_D)}+\frac{\mathbb{E}_0\left[\mathrm{CF}_2\right]}{(1+r_D)^2}+\frac{\mathbb{E}_0\left[\mathrm{CF}_3\right]}{(1+r_D)^3}+\ldots+\frac{\mathbb{E}_0\left[\mathrm{CF}_T\right]}{(1+r_D)^T}-\mathrm{Price}_0=0$$

- General Rule : (Promised) Yield to Maturity $\geq$ Cost of Debt
  - Inequality is strict iff there is any probability of default with less than 100% recovery:
    - The difference between the yield to maturity and the cost of debt increases with the probability of default
    - Yield to Maturity $-$ Cost of Debt $=$ Expected Default Loss

The Cost of Capital (WACC) — Weights

Università Bocconi
DEPARTMENT OF FINANCE



Cost of Capital: Weights for Computation

General rule: compute weights using **market value weights** (not book value weights)

Reason: the cost of capital measures the cost of issuing securities (debt, equity, etc.), which are issued at market value, not at book value

- **Debt** (net of cash & cash equivalents):
  - Only interest bearing debt, not all liabilities (exclude trade payable, ...), both short and long term
  - Capitalize PV of future operating lease payments: we have to consider off-balance sheet Debt
  - Estimate Debt Value of non-traded Debt:

$$D_{\text{Market Value}} = \frac{\text{Coupon}}{r_d}\left[1 - \frac{1}{(1+r_d)^t}\right] + \frac{\text{Face Value}}{(1+r_d)^t}$$

  - Modify interest coverage (IC) ratio when estimating synthetic rating:

$$\text{Modified ICR} = \frac{(\text{EBIT} + \text{Operating Lease Expenses})}{\text{Interest Expenses} + \text{Implicit Interest Expenses on Leases}}$$

- **Equity**:
  - Current stock price $\times$ # shares outstanding; if company is private, use comps' multiples
  - Need to include all equity claims (minorities, warrants, management stock options, ...)
- **Hybrid Securities**:
  - Need to include also their market value (and required rate of return)

The Cost of Capital (WACC) — Recap

Università Bocconi
DEPARTMENT OF FINANCE



Cost of Capital: Recap

Tax Rate

Marginal Tax Rate reflecting tax benefits of Debt

Market Value

1. Debt
2. Equity
3. Hybrid Securities

$$WACC = r_d(1 - t_c)\frac{D}{(D+E+S)} + r_e\frac{E}{(D+E+S)} + r_s\frac{S}{(D+E+S)}$$

Cost of Debt (r_d)

Based of:
- current cost of borrowing
- synthetic rating

Cost of Equity (r_e)

refer to previous details

Cost of Hybrid Securities (r_s)

It has to reflect expected payments and current market values of any hybrid securities

Takeaways

- In general, cost of debt $k_d \leq$ promised yield, with equality iff default probability is zero. We outlined different methods for estimating k_d:
  - Equate the sum of PV of expected cash flows to current debt value and solve for the IRR
  - Use a CAPM approach that adds a risk premium to AA yields as a function of rating class
  - Use credit default swaps as a guide
  - If everything else fails, use comparables and examine their cost of debt
- The WACC is used to value the firm when it operates with a fixed leverage, $\frac{D}{D+E}$
- Financial distress costs and agency costs are countervailing forces to the tax advantages of debt financing